Local Topological Fragility in Game of Life and HighLife

Kunal Bhatia

Independent Researcher, Heidelberg, Germany[a)]

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Life-like cellular automata exhibit *local topological fragility:* the initial density of weakly-connected live cells predicts future loss of fine-scale connected components. Across Conway's Game of Life (GoL) and HighLife, two grid sizes ($L \in \{64, 128\}$), four densities, and seven forecast horizons, a sixteen-statistic topology battery identifies a predictive hierarchy: initial component count $C(0)$ (mean residual $R^2 = 0.352$), small-component counts (0.330), singleton/orthogonal-degree-zero cells (0.274), and *embedded isolates*—alive cells with no orthogonal and at least one diagonal live neighbour—as the mechanistically transparent motif ($R^2 = 0.234$; prestate $R^2 = 0.241$; min 0.175 at $k = 200$; shuffle ≈ 0.0008).

The effect is target-specific: both shuffle increments are near zero ($\Delta R^2 < 0.001$), and the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$; CV $\lesssim 0.17$). Mechanistically, any orthogonal-degree-zero cell with Moore degree 0 or 1 dies at $t+1$ as an exact S_{23} rule consequence; multi-diagonal isolates more often survive-connected. Both pathways reduce the component count; a local-window feature captures this ($R^2 = 0.538$). The standardised mechanism transfers across held-out conditions ($R_z^2 = 0.544$), and raw amplitude is calibrated from (L, ρ) (LOO $R^2 = 0.970$).

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response, mechanism transfer, observer-dependent description

[a)]https://orcid.org/0009-0007-4447-6325

I. INTRODUCTION

Conway's Game of Life (GoL)[1] and HighLife are among the most studied cellular automata, valued for their deterministic dynamics, rich emergent structure, and computational tractability. Earlier work established edge-of-chaos properties[2,3] and statistical complexity[4] of Life-like rules, and identified the role of coherent structures[5] and computational irreducibility[6] in deterministic CAs. Local causal-state methods[4,5] identify predictive-equivalence partitions from past/future lightcones; our work instead asks which simple hand-defined structural classes are privileged for a specific coarse-grained target observable. The partial predictability of a chosen observable demonstrated here is consistent with computational irreducibility[6]: we do not claim to predict the full future CA configuration, only one low-dimensional target statistic.

We identify *local topological fragility*—the initial density of structurally isolated or weakly-connected live cells—as a target-indexed property of Life-like CA states that predicts future fine-component loss. A battery of sixteen topology statistics reveals a predictive hierarchy: initial component count $C(0)$ is the strongest aggregate predictor; small-component and singleton-component (orthogonal-degree-zero) counts are the strongest local-structure predictors; and the count of *embedded isolated cells* (iso_count)—alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour—is the most analytically transparent motif within that family. Embedded isolates sit at the topological boundary of 4-connectivity: orthogonally isolated, diagonally tethered, they are the mechanistically cleanest fragile connectivity nodes in the grid, because their one-step fate under the S_{23} rule has a simple analytic form. Earlier observer-scale diagnostics from the same research program motivated this target-specific analysis; they are documented in the public repository.

The main results, established across GoL and HighLife, two grid sizes ($L \in \{64, 128\}$), four density bands ($\rho \in \{0.20, \ldots, 0.35\}$), and seven prediction horizons, are: (1) a topology-baseline battery of 16 statistics shows that local fine-topological fragility—led by $C(0)$, small-component counts, and singleton/orthogonal-degree-zero counts—strongly predicts fine-net change, with embedded isolates as the most analytically transparent motif; (2) iso_count predicts future fine-component loss with mean prestate $R^2 = 0.241$, sustained above 0.175 at $k = 200$; (3) the effect is target-specific—incremental lift for density, entropy, and coarse-block targets is near zero, as are both shuffle nulls; (4) the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative

in all 112 condition-horizon combinations, with CV $\lesssim 0.17$; (5) the mechanism is local component-context loss, not simple cell death, and transfers across all leave-one-out axes.

Figure 1 provides the entry point: the structural definition of an embedded isolate and the target-specific ΔR^2 pattern that singles out the fine-component target.

II. DEFINITIONS AND PROTOCOL

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a). We denote the 4-connected component count at time t as $C(t)$ and define the fine-net change at horizon k as $\Delta C_k = C(t+k) - C(t)$.

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons $\times$ 16 conditions = 112 condition-horizon slope estimates.

The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares (OLS) with $N_{\text{live}}(0)$ (total live-cell count, proportional to ρ) as a nuisance covariate, so that iso_count is evaluated for its density-orthogonal contribution. This and all horizon-response regressions are fitted *condition-wise* (within each of the 16 batteries independently); the global $\bar{\beta}_{\text{iso}}$ reported in the text is the mean across conditions. Shuffle-null analyses (iso-count label permutation or target permutation) are also run within condition. Mechanism R^2 values are 5-fold cross-validated within condition. Slope confidence intervals use world-resampling bootstrap (≥ 300 draws per condition) rather than parametric OLS standard errors. Each initial grid is drawn independently, making worlds the natural bootstrap unit.

We use four distinct R^2 quantities throughout and distinguish them as follows. (a) *Ordinary R^2*: variance explained by an OLS model with no cross-validation; used for prestate fits.

(b) *Residual* R^2: R^2 of iso_count after the density covariate has been projected out; this is the within-condition prestate R^2 reported in Section III and Table I. (c) *Cross-validated* R^2 (5-fold, within condition): used for mechanism features in Section V. (d) ΔR^2: the difference $R^2(\text{density + predictor}) - R^2(\text{density only})$; used in the target-selection screen (Section III) to measure incremental lift over the density baseline.

All results are scoped to these two Life-like rules on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Low-density dynamics ($\rho \lesssim 0.10$), other Life-like rules, and non-random initial conditions are outside the present scope. We do not rerun experiments; all statistics are drawn from the fixed set of CSVs in `outputs/`.

III. TARGET-SPECIFIC SELECTION AND NON-LEAKY PRESTATE

A. Target-specific ΔR^2

Figure 1b shows the selection-principle battery: for each candidate predictor we measure $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$ across six displayed targets, with iso_count, coarse block-variance, block entropy, and two shuffle nulls as predictors (see Appendix A for the residualisation protocol).

iso_count on the fine-net target yields $\Delta R^2 = 0.0012$, a positive incremental lift in both GoL and HighLife screens and in the global pooled screen. Both shuffle nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not an artefact of the density-count correlation. iso_count adds essentially nothing for density, block-variance, block-entropy, or coarse-component targets: the uplift is concentrated on the fine-net target only. This specificity rules out explanations based purely on total alive-cell count or generic spatial statistics.

Two distinct R^2 *quantities.* $\Delta R^2 = 0.0012$ and the prestate $R^2 \approx 0.241$ reported in the next subsection answer different questions and should not be compared directly. The ΔR^2 screen asks whether iso_count adds *selective* information beyond the live-cell-count baseline *across multiple targets simultaneously*; because the baseline already absorbs most of the pooled variation, the incremental lift is numerically small by construction. The prestate R^2 evaluates the density-residualised fine-component target directly within each condition–horizon battery: how much of the remaining variation does $t = 0$ iso_count explain? That is the main effect-size estimate—iso_count accounts for roughly 0.241 of residual

fine-component-change variation on average across horizons, falling no lower than 0.175 at $k = 200$.

B. Non-leaky prestate recovery

A natural concern is whether iso_count is a proxy for information that requires knowing the one-step fate of individual cells. Figure 2 settles this: using *only* iso_count at $t = 0$—no fate features, no local-window information—the prestate model achieves mean $R^2 = 0.241$ across all horizons $k \in \{1, \ldots, 200\}$ (minimum $R^2 = 0.175$; shuffle ≈ 0.0008). Adding the full diagonal-class breakdown gives no substantial gain. The full multi-horizon iso-count signal is recoverable from the t=0 count alone.

IV. TEMPORAL RESPONSE CURVE

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

All 112 slopes are negative; all 112 bootstrap slope confidence intervals are negative (= 1.00 at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$ (range ≈ 0.09) while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains well above the null at every horizon.

Figure 3 shows the slope and R^2 curves. The slope plateau from $k = 5$ to $k = 200$ indicates that the mechanism acts primarily in the first step: once the initial fragile nodes are resolved at $t = 1$, the remaining dynamics do not strongly modulate the iso_count signal. The CV profile confirms robustness: condition-level spread is small relative to the global mean, and GoL and HighLife track each other closely despite different neighbourhood rules.

Figure 4 provides the complete picture: a 16×7 heatmap of all condition-horizon slopes, every cell shaded negative. There are no outlier conditions or horizons.

V. MECHANISM: LOCAL COMPONENT-CONTEXT LOSS

Having established *that* iso_count predicts fine-component change, we turn to *how* this occurs—what actually happens at $t \to t + 1$ when embedded isolates reduce the future

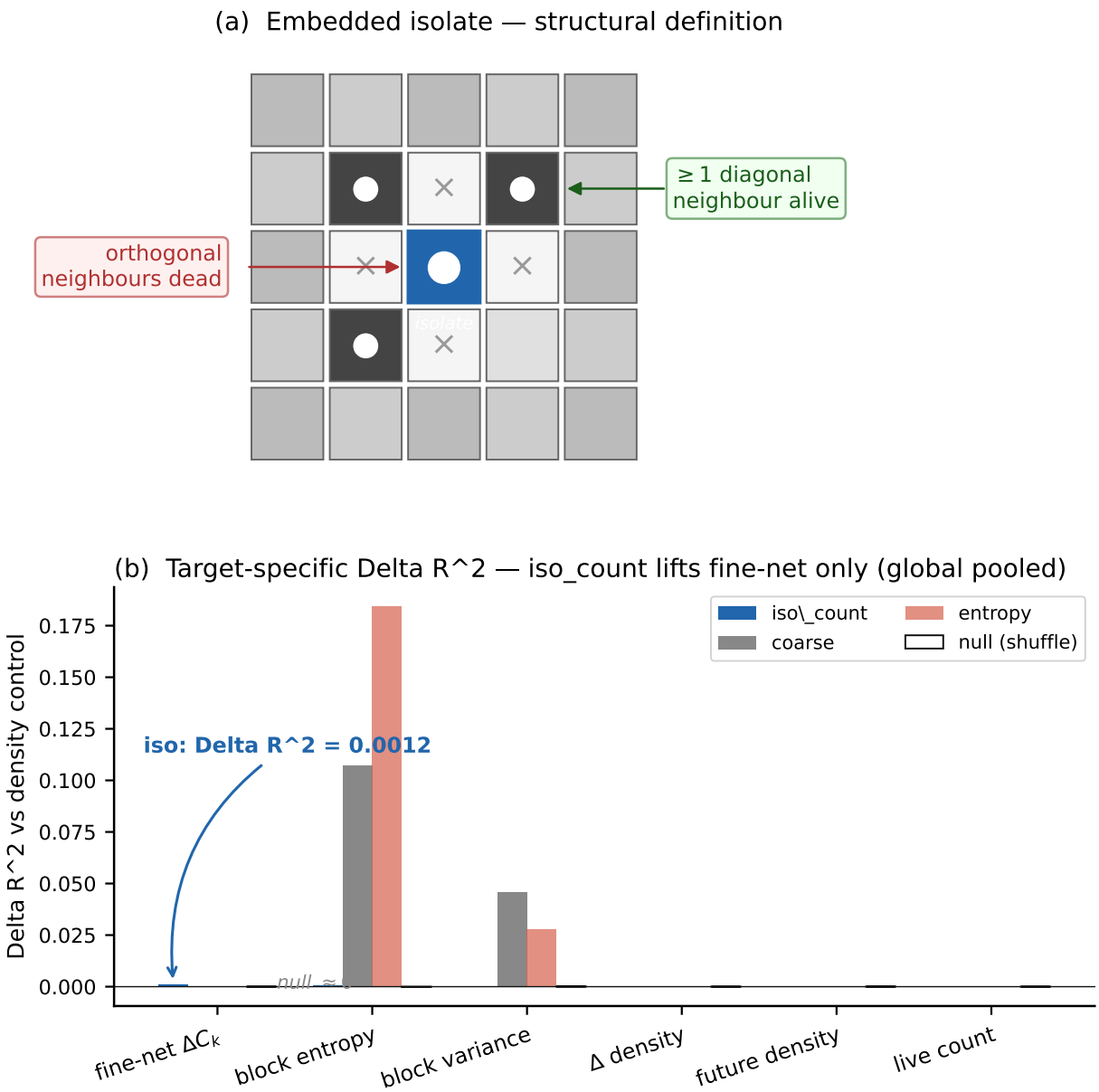

FIG. 1. **Structural definition and target-specific selection.** (a) An embedded isolate: an alive cell (filled circle) with all four orthogonal neighbours dead (×) and at least one diagonal neighbour alive (filled grey). This configuration places the cell at the most fragile locus of the local component topology. (b) ΔR^2 bars for iso_count and comparison predictors (coarse, entropy, null) on six displayed targets (global pooled; `target_delta_components` is numerically identical to fine-net and omitted). iso_count adds a positive incremental $\Delta R^2 = 0.0012$ for fine-net only; both nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not a density artefact.

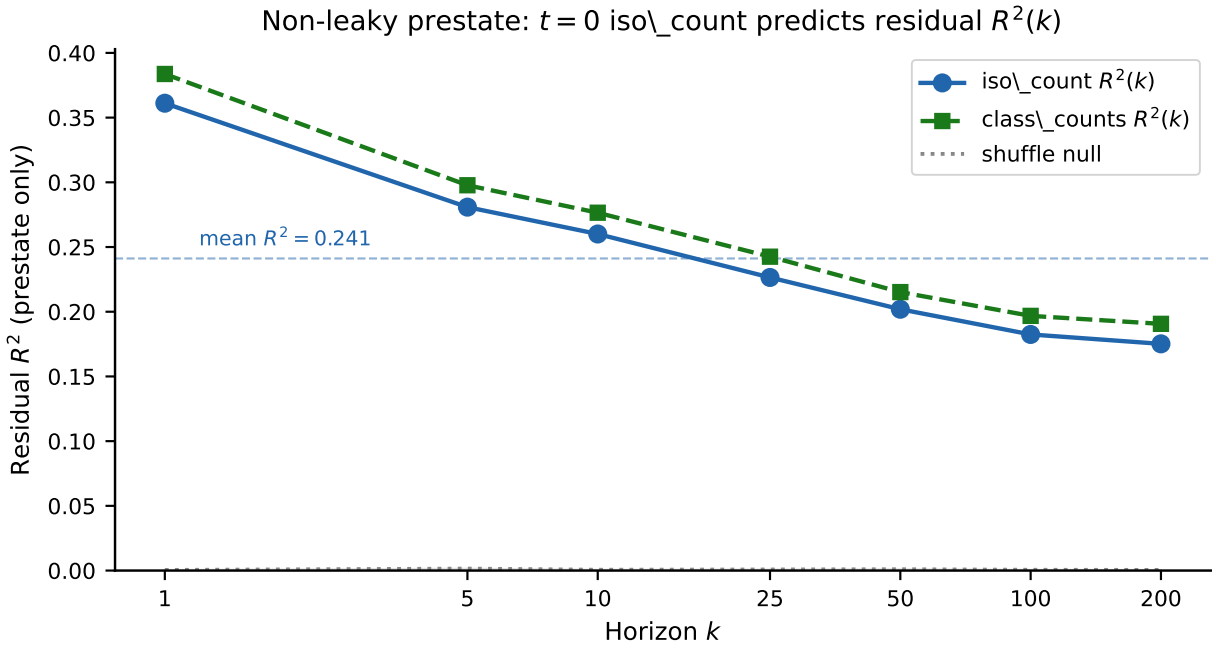

FIG. 2. **Non-leaky prestate recovery.** Using only $t = 0$ iso_count (no fate features, no local-window information), residual $R^2(k)$ stays above 0.175 at every horizon and averages 0.241 (dashed line). Adding class-count breakdown (green) gives marginal improvement. The grey band shows the shuffle null ($R^2 \approx 0.0008$). The tested iso-count signal is recoverable from the t=0 statistic alone.

TABLE I. Temporal response summary (16 conditions, 1000 worlds each; bootstrap CIs, ≥300 draws). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

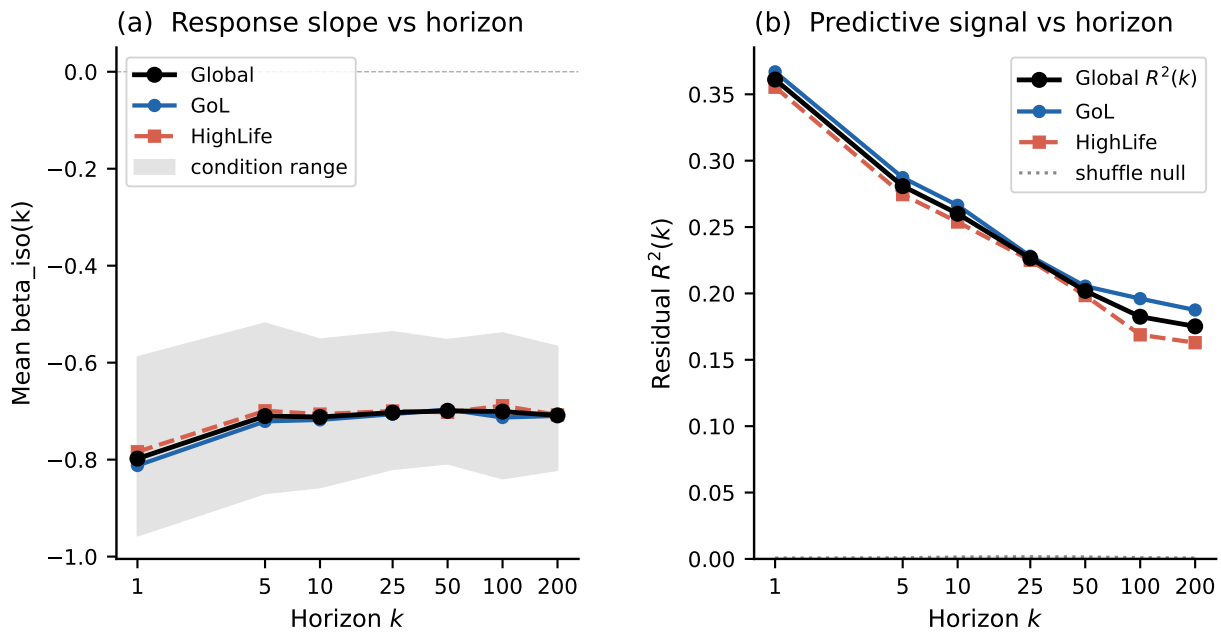

FIG. 3. **Temporal response: slope and predictive signal.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (blue circles), HighLife (orange squares), and global mean (black); shaded region shows condition-level spread. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; grey band marks the shuffle null. Predictive power decays gracefully but remains far above null at all tested horizons.

component count?

One-step transition taxonomy. At the one-step level, each embedded isolate can affect local component count via four transition types: (i) *focal death*: the isolate cell dies, removing a local connectivity node; (ii) *survival-connected merger*: the cell survives and fuses with an adjacent component, absorbing an independent fine component and reducing the count by one; (iii) *birth-mediated bridge*: a newly born neighbour cell connects the isolate to another component; and (iv) *local gain/fragmentation*: births around the isolate create new independent components, partially offsetting the loss. Type (i) is partially analytic: for single-diagonal isolates the focal cell has exactly one live Moore-neighbour, so it dies under S_{23} regardless of surrounding context (see below). Types (ii)–(iv) depend on the surrounding 5×5 neighbourhood configuration and are measured empirically.

We decompose iso_count into one-step outcome categories: death, survival, survival-connected (merges with a larger component), or triggering births at orthogonal or diagonal

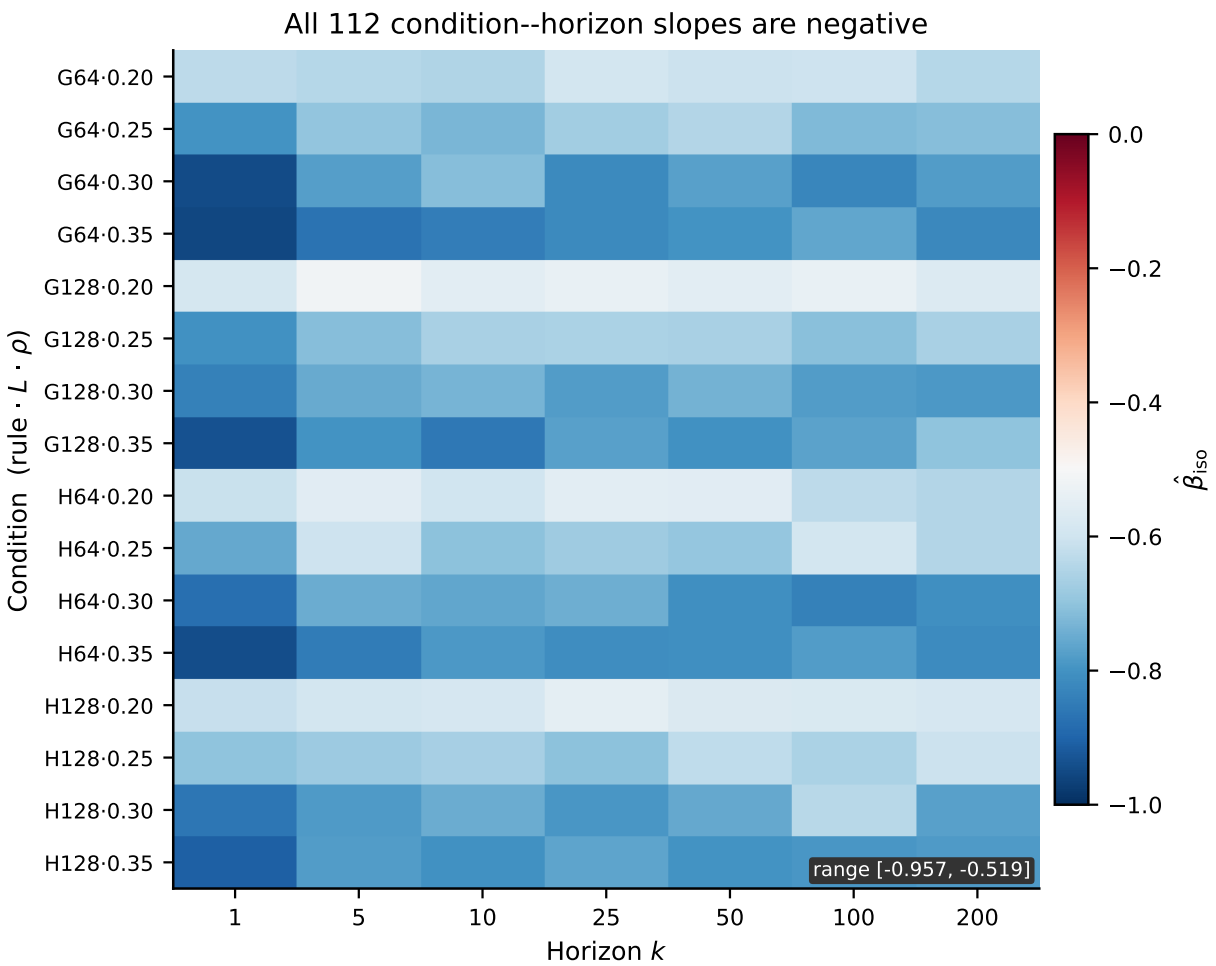

FIG. 4. **All 112 condition–horizon slopes are negative.** Full 16×7 heatmap of $\hat{\beta}_{\text{iso}}$. Rows are conditions (rule $\cdot$ L $\cdot$ ρ), columns are horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$. Every cell is negative, confirming the negative temporal response without exception.

sites. We also compute a *local-window* feature: the total component-count decrease within a 5×5 window centred on each isolate over one step. Cross-validated R^2 values for these features are shown in Fig. 5a.

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net mechanism is *local component-context loss*, not simple cell death. Figure 5b shows the standardised slopes for each fate feature, revealing the partial cancellation between loss and gain.

Class-level analysis (Fig. 6) reveals a consistent physical story.

a. Exact deaths among orthogonal-degree-zero cells. Any alive cell with orthogonal degree 0 has Moore-neighbourhood count equal to its diagonal-neighbour count. Under S_{23}, survival requires 2 or 3 live Moore-neighbours; therefore: truly isolated cells (Moore degree 0, no live neighbours at all) die; single-diagonal embedded isolates (classes 1, 2, 4, 8; Moore degree 1) die; and four-diagonal isolates (class 15; Moore degree 4) also die. All three cases are exact consequences of the transition table, independent of surrounding context; death fractions of 1.0 for these classes (Fig. 6) are analytically guaranteed. Two- and three-diagonal isolates (Moore degrees 2 and 3) survive under S_{23} and depend on the surrounding 5×5 context for component merging. This analytic structure directly explains why the orthogonal-degree-zero superset outperforms the embedded-isolate subset in raw predictive R^2 (Table III): it combines both dying subclasses (Moore degree 0 and 1) under a single count. Embedded isolates remain the mechanistically preferred motif because the diagonal-tethering condition gives them a concrete component-context interpretation absent from truly isolated floating cells.

Two- and three-diagonal embedded isolates more often survive-connected: they merge with a larger component and reduce the fine component count by absorbing an independent component. Both pathways lead to the same aggregate outcome—local fine-connectivity loss—which is why the local-window loss feature ($R^2 = 0.538$) dominates over simple death counts ($R^2 = 0.355$ for iso_count alone). The dominant process is the disappearance of fine-net connectivity in the local neighbourhood, aggregated over many isolate classes and many conditions.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

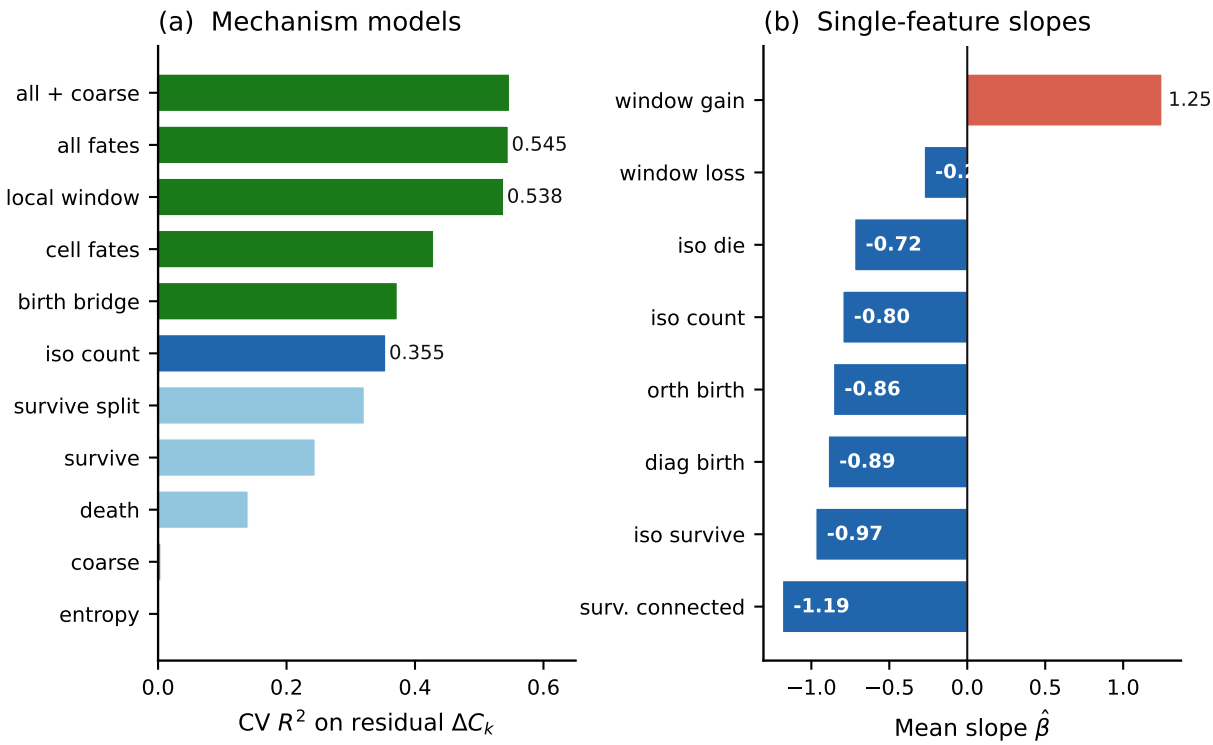

FIG. 5. **Mechanism carrier decomposition.** (a) Cross-validated R^2 for each fate and mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features are negative, gain-related features are positive, revealing partial cancellation.

VI. ROBUSTNESS: TRANSFER, AMPLITUDE, AND LOCAL BASELINES

The mechanism analysis tells us what the carrier is. Two robustness questions remain: does it generalise to held-out conditions, and does the scale of the response follow a systematic pattern?

A. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). All four transfer axes pass (Fig. 7a). For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive $= 1.00$ on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$). The standardised mechanism

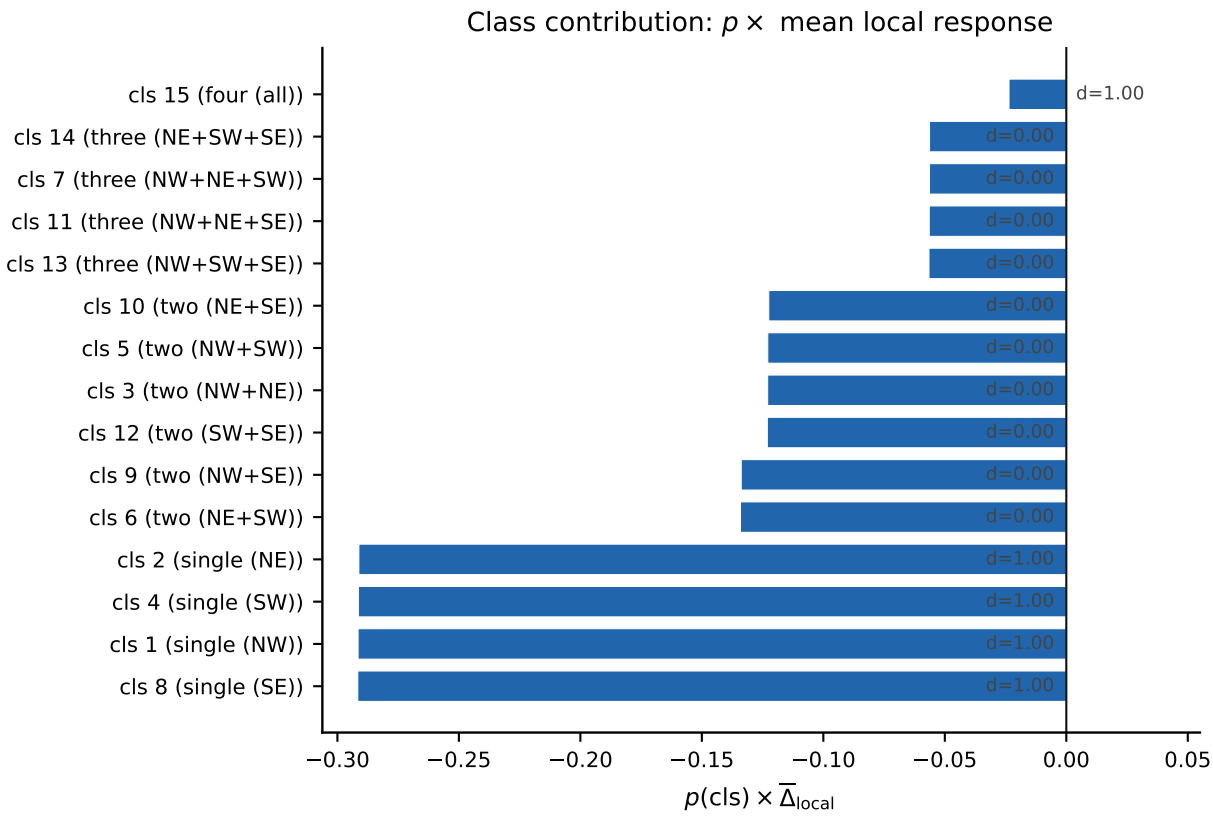

FIG. 6. **Class-level contributions to the mechanism.** Horizontal bars show $p(\text{cls}) \times \overline{\Delta}_{\text{local}}$ for each diagonal-adjacency class. Annotations give the death fraction (d) from GoL transition-class statistics. All classes contribute negatively; single-diagonal-neighbour classes drive the bulk via their high prevalence.

direction is condition-invariant: whichever leave-one-out split is applied, a model trained on the remaining conditions predicts the held-out condition well.

B. Amplitude Calibration

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV = 0.369) but are predictable from grid size and density. The parsimonious size+ρ model—two predictors, centred $\log_2 L$ and centred ρ—achieves LOO $R^2 = 0.970$ on the 16 condition-level amplitudes. The full model (six predictors: centred $\log_2 L$, centred ρ, rule indicator, and three cross-terms) adds only marginal gain (LOO $R^2 = 0.977$). Size-only gives LOO $R^2 \approx 0.802$; rule-only gives -0.138, confirming that grid size L is the dominant predictor and rule iden-

tity alone is not. Because the model is fit at the 16-condition level, these calibration results should be read as within-grid evidence rather than a universal scaling law.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L,\rho)\,\hat{\beta}_{\text{iso},z}\,, \tag{1}$$

where $\hat{\beta}_{\text{iso},z} \approx -0.50$ is the transferable standardised slope and $A(L,\rho)$ is a condition amplitude calibrated from (L,ρ) with LOO $R^2 = 0.970$ for the two-predictor model. The natural explanation is that L sets the number of components and hence the scale of fine-component change, while ρ controls the density of the structural class; together they set the amplitude scale that multiplies the transferable standardised direction.

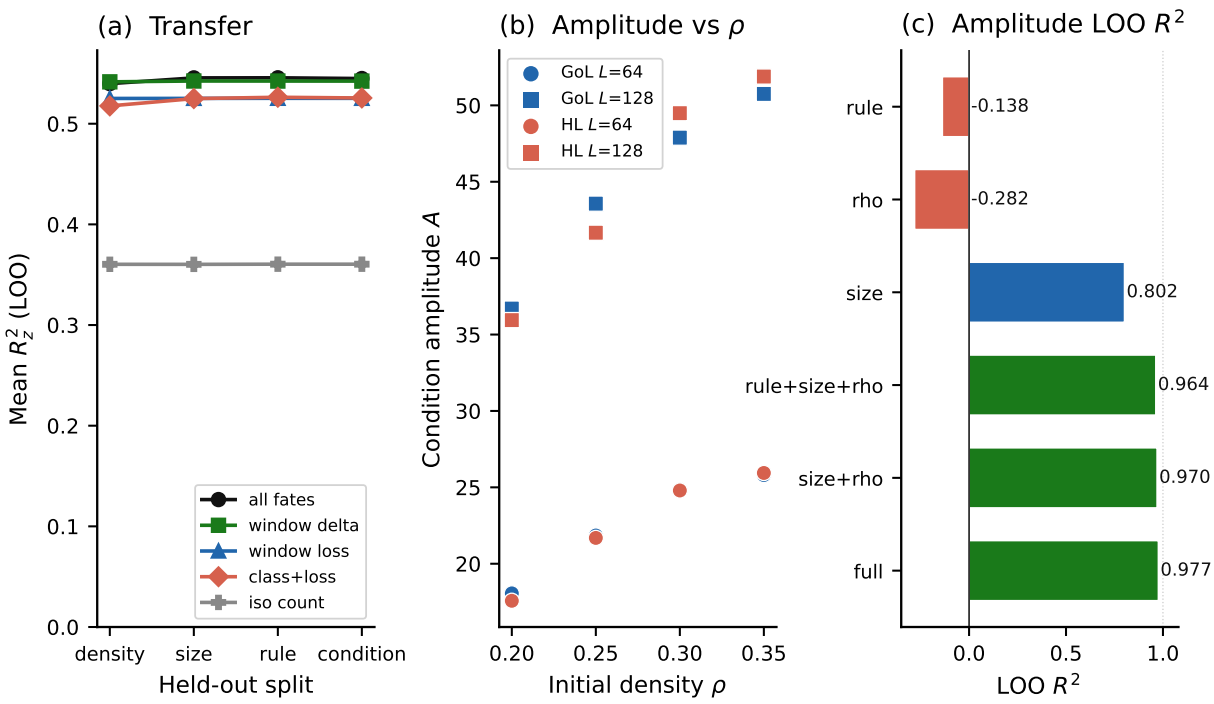

FIG. 7. **Transfer and amplitude calibration.** (a) Standardised transfer R_z^2 for five mechanism models across four leave-one-out axes; all lines stay positive. (b) Condition amplitude versus initial density ρ, coloured by grid size L and rule. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$, size+ρ alone reaches 0.970.

C. Local Topology Baselines

Because the target is a change in 4-connected component count, embedded isolates might be merely one member of a broader family of local topological fragility statistics, each predicting ΔC_k for similar reasons. We test this directly by running the same density-residualised protocol (Appendix A) on fifteen alternative t=0 statistics: initial component count $C(0)$, small-component counts (sizes ≤ 4, $= 2, 3, 4$), singleton components (algebraically identical to orthogonal-degree-zero cells), truly isolated cells (Moore degree 0), Moore-degree counts ($d = 1, 2$), boundary cells (alive with ≥ 1 orthogonally dead neighbour), and coarse block statistics. Table III shows global mean residual R^2 per predictor, averaged over all 112 condition-horizon cells.

The result is structurally informative and transparent. $C(0)$ is the strongest predictor ($R^2 = 0.352$): more components at t=0 means more components "at risk" of loss. Small components of size ≤ 4 are next ($R^2 = 0.330$), followed by singleton components / orthogonal-degree-zero cells ($R^2 = 0.274$). Embedded isolates rank fourth ($R^2 = 0.234$), behind the orthogonal-degree-zero superset from which they are drawn. All block-level and Moore-degree-only statistics are substantially weaker.

Embedded isolates are therefore not claimed to be the unique or globally optimal predictor. They are a compact, diagonal-tethered subclass of the orthogonal-degree-zero family, singled out because their one-step fate has a clean analytic form under S_{23} and because the diagonal-tethering condition provides a concrete component-context interpretation: a cell touching live structure diagonally but not orthogonally is a fragile link whose death or merger directly removes a fine-connectivity node. The broader local-fragility family—from $C(0)$ to small-component counts to singleton cells to embedded isolates—forms a coherent predictive structure that this paper characterises and places on a mechanistic footing.

VII. DISCUSSION

Summary. Local topological fragility—the initial density of weakly-connected live cells—predicts future fine-component loss in GoL and HighLife. The topology-baseline battery (Table III) places this on a tested footing: the predictive hierarchy from $C(0)$ through small-component counts and singleton/orthogonal-degree-zero cells to embedded isolates

TABLE III. Local topology baseline: mean residual R^2 per predictor, averaged over all 112 condition-horizon cells (2 rules × 2 sizes × 4 densities × 7 horizons; 1000 worlds per condition). All predictors measured at t=0; live-count + density baseline residualised before fitting. Singleton components ≡ orthogonal-degree-zero cells by definition.

Predictor	Mean resid. R^2
$C(0)$ (initial component count)	0.352
Small components (≤ 4 cells)	0.330
Singleton / orth-degree-zero cells	0.274
Embedded isolates (iso_count)	**0.234**
Moore-degree-one cells	0.033
Block variance / entropy	≈0.006–0.009

($R^2 = 0.234$) forms a coherent fragility family, with embedded isolates as the mechanistically transparent motif. The target-specific prestate signal ($R^2 = 0.241$; both shuffle nulls near zero) arises from a local mechanism—orthogonal-degree-zero cells with Moore degree 0 or 1 die exactly under S_{23}—that standardises and transfers across all leave-one-out axes, with raw amplitude calibrated from (L, ρ) (LOO $R^2 = 0.970$).

Common-direction diagnostic. A descriptive check on why a single statistic serves all fine-net horizons is to measure alignment of regression-coefficient vectors in feature space. For each task we fit a residual model and normalise the coefficient vector; pairwise absolute cosines $|\cos\theta_{ij}|$ measure task alignment. Fine-net horizon tasks are near-collinear ($|\cos| = 0.999$, rank-1 energy = 0.999): a single feature direction aligns all k-step predictions simultaneously. Heterogeneous targets are not ($|\cos| = 0.427$), and no single predictor adds meaningful ΔR^2 for all of them. Figure 8 shows this contrast. This collinearity is a descriptive geometric property of the regression directions, not a claim about any specific theory of causal emergence or effective coarse-graining.

The two-layer structure. The decomposition $\hat{\beta}_{\text{iso}}(\text{cond}) \approx A(L, \rho)\, \hat{\beta}_{\text{iso}}, z$ cleanly separates the transferable standardised mechanism from the condition-specific scale. This matters for applications: if a practitioner has access to a new (L, ρ) condition, they can predict the raw slope amplitude with LOO $R^2 \approx 0.970$ from (L, ρ) alone, and they can apply the standardised slope direction directly from the training pool. The two layers are independently portable.

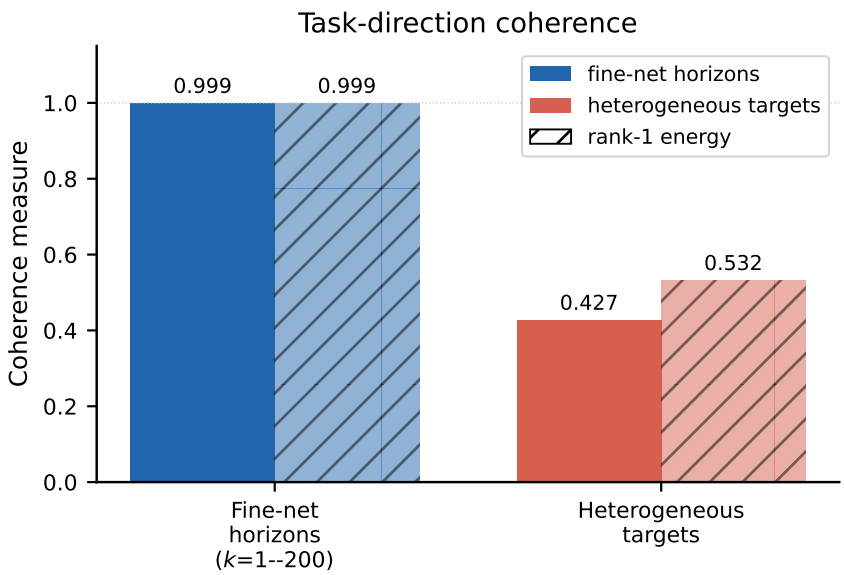

FIG. 8. **Task-direction coherence.** Mean pairwise $|\cos\theta|$ (solid) and rank-1 energy (hatched) for fine-net horizon tasks (blue) and heterogeneous targets (orange). Fine-net horizons are near-collinear ($|\cos| = 0.999$, rank-1 $= 0.999$); heterogeneous targets are dispersed ($|\cos| = 0.427$).

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Relation to structural and computational work. Local causal-state methods[4,5] infer predictive-equivalence partitions from joint past/future lightcone statistics. Embedded isolates are a hand-defined structural class within that broader space: they are not claimed to be a full predictive-equivalence partition, but they capture one target-specific component of future fine-connectivity change with a simple geometric definition. The amplitude calibration does not conflict with coarse-graining impossibility results[6]: we predict one low-dimensional target statistic (ΔC_k), not the full future CA configuration. The partial predictability shown here is consistent with computational irreducibility; it concerns a specific chosen observable, not global state reconstruction. The novelty lies in combining four elements simultaneously: a coherent local-fragility family that (i) is characterised by a base-

line battery rather than a single hand-picked object, (ii) has a mechanistically transparent motif (embedded isolates) with exact one-step fate for the single-diagonal subclass, (iii) produces a standardised mechanism that transfers across held-out conditions, and (iv) admits amplitude calibration from (L, ρ)—all for future fine-component loss in two well-studied Life-like rules.

Local topological fragility as an observer class. The topology-baseline battery shows that what matters for predicting fine-component change is the initial density of locally disconnected or weakly-connected live cells—a property we call local topological fragility. $C(0)$, small-component counts, and orthogonal-degree-zero cells all capture this property in different ways and form a coherent predictive family. Embedded isolates are the analytically cleanest member of that family: their diagonal tethering gives them a component-context interpretation (they are nodes at risk of losing their local connectivity), and their one-step fate is partially analytic rather than purely empirical. The stability of $\hat{\beta}_{\text{iso}}(k)$ across horizons suggests that the risk resolves primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result. iso_count is therefore a *topological risk indicator* within the broader local-fragility family, not a simple density proxy and not claimed to be the uniquely optimal fragility statistic.

VIII. CONCLUSION

Future fine-component change in Conway's Game of Life and HighLife is controlled by local topological fragility: the initial density of weakly-connected live cells. A topology-baseline battery of sixteen statistics identifies a predictive hierarchy from the initial component count $C(0)$ ($R^2 = 0.352$) through small-component and singleton/orth-degree-zero counts ($R^2 = 0.330$, 0.274) to embedded isolates ($R^2 = 0.234$), the most analytically transparent motif. The $t = 0$ embedded-isolate count alone explains a substantial fraction of residual fine-component change across all tested horizons (mean prestate $R^2 = 0.241$; minimum $R^2 = 0.175$ at $k = 200$; shuffle ≈ 0.0008). The residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations (7 horizons $\times$ 16 conditions; 2 rules $\times$ 2 sizes $\times$ 4 density levels), an empirical regularity that holds without exception across the scoped battery. Both iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The one-step mechanism is local component-context loss, not

simple cell death: any orthogonal-degree-zero cell with Moore degree 0 or 1 dies exactly under the shared S_{23} rule; two- and three-diagonal isolates often survive-connected. This standardised mechanism transfers across all leave-one-out axes and the raw amplitude is predictable from (L, ρ) (LOO $R^2 = 0.970$, rising to 0.977 in the full six-predictor model).

The collinearity of fine-net regression directions ($|\cos| = 0.999$) provides a compact geometric diagnostic of why one prestate statistic serves all tested horizons simultaneously.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`. The repository includes an automated regression suite (93 tests) covering the selection, horizon-response, mechanism, transfer, amplitude-calibration, and artifact-generation pipelines, plus a standalone topology-baseline controls script (`ca_topology_baseline_controls.py`) whose outputs are archived in `outputs/topology_baselines/`; all regression tests pass for the submitted build.

Appendix A: Regression Details and Null Definitions

Residualisation protocol

For each prediction world, the density nuisance covariate is $N_{\text{live}}(0)$, the total live-cell count at $t = 0$ (proportional to ρ up to Bernoulli sampling variance). All regression models include $N_{\text{live}}(0)$ as a linear covariate before adding iso_count or any fate feature. $\Delta R^2 = R^2(\text{density} + \text{predictor}) - R^2(\text{density only})$. The residual slope $\hat{\beta}_{\text{iso}}$ is the OLS coefficient on iso_count in this density-residualised fit.

Null definitions

Two nulls are used: (i) *iso-shuffle null*: iso_count labels are permuted uniformly at random within each condition; ΔR^2 and slope are recomputed. (ii) *target-shuffle null*: the target ΔC_k is permuted; ΔR^2 is recomputed with the original iso_count. Both converge near zero ($\Delta R^2 < 0.001$), confirming the signal depends on the joint alignment of iso_count with ΔC_k and is not a density or permutation artefact.

Appendix B: Transition-Class Coding

Diagonal-adjacency class is an integer in $[1, 15]$ indexing the bit pattern of the four diagonal neighbours of an embedded isolate (NW, NE, SW, SE), with class c equal to the binary value $NW \cdot 1 + NE \cdot 2 + SW \cdot 4 + SE \cdot 8$. Class 15 (all four diagonal neighbours alive) is rare in this battery ($p \approx 0.009$, $n = 80{,}015$ instances) but contributes a negative local response ($\bar{\Delta} \approx -2.52$, death fraction $= 1.0$) comparable to other multi-diagonal classes; it is not excluded from the analysis. Classes 1, 2, 4, 8 are the *single-diagonal* classes; classes 3, 5, 6, 9, 10, 12 are the *two-diagonal* classes; classes 7, 11, 13, 14 are the *three-diagonal* classes; class 15 is the *four-diagonal* class.

Appendix C: Background Observer-Scale Diagnostics

The main text establishes the embedded-isolate temporal response using the 16-condition GoL/HighLife battery described in Section II. For lineage and context, this appendix records earlier observer-scale diagnostics from the same research program. These diagnostics show *why* a target-indexed description analysis was a natural next step: fine and coarse observers disagree systematically, and different targets are optimally predicted at different descriptive scales. **These figures are background diagnostics only; they are not load-bearing evidence for the response estimates in the main text.**

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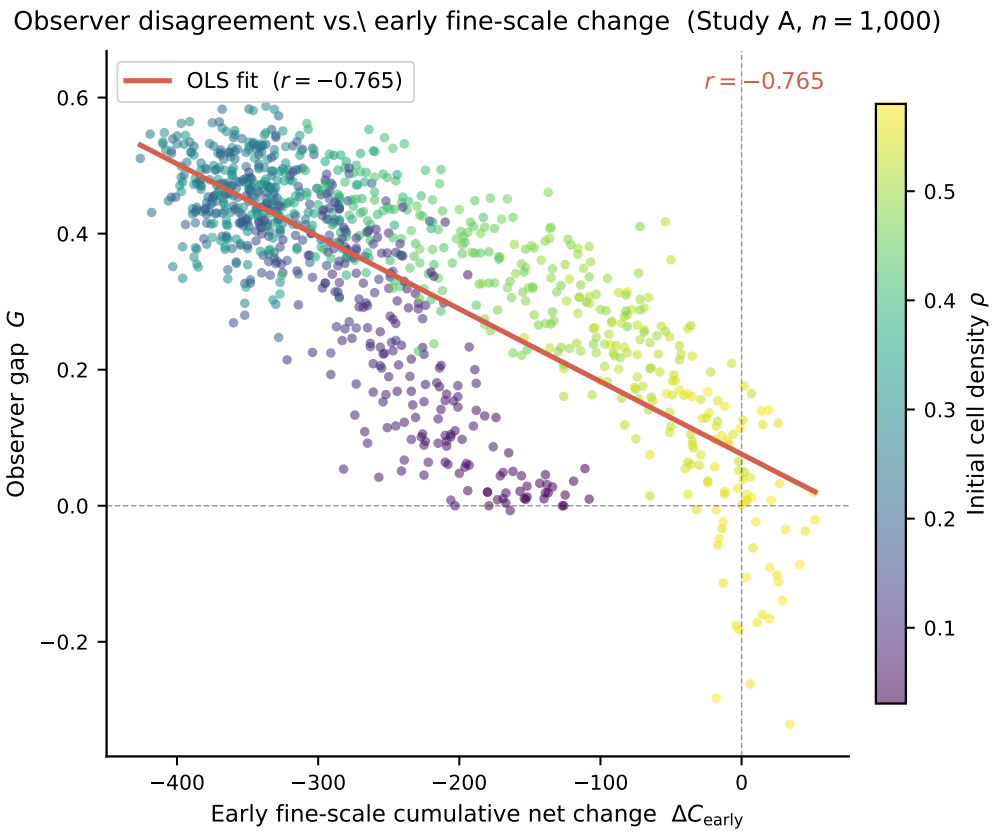

FIG. 9. **E1 — Observer disagreement scatter (Study A, $n = 1{,}000$).** Each point is one simulated GoL world. x-axis: early fine-scale cumulative net component change ΔC_{early}; y-axis: observer gap G (fine minus coarse observer agreement); colour: initial cell density ρ. The regression line gives $r = -0.765$ ($p \ll 10^{-100}$); partial r controlling for density $= -0.848$. Worlds where fine-scale structure degrades early tend to produce larger observer disagreement. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

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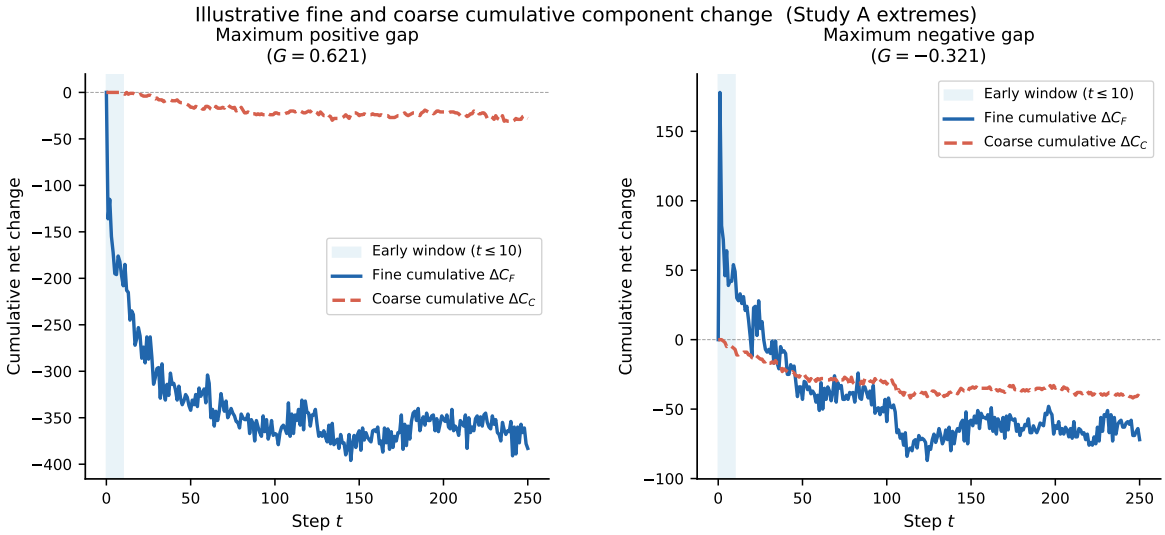

FIG. 10. **E2 — Illustrative fine and coarse observer trajectories (Study A extremes).** Left: the world with the maximum positive observer gap ($G = +0.621$). Right: the world with the maximum negative gap ($G = -0.321$). Blue solid: fine-scale cumulative net component change $\Delta C_F(t)$. Orange dashed: coarse-scale cumulative net change $\Delta C_C(t)$. Shaded region: early window ($t \leq 10$) whose summary statistic ΔC_{early} is used in Figure 9. These are selected extreme cases, not representative samples. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

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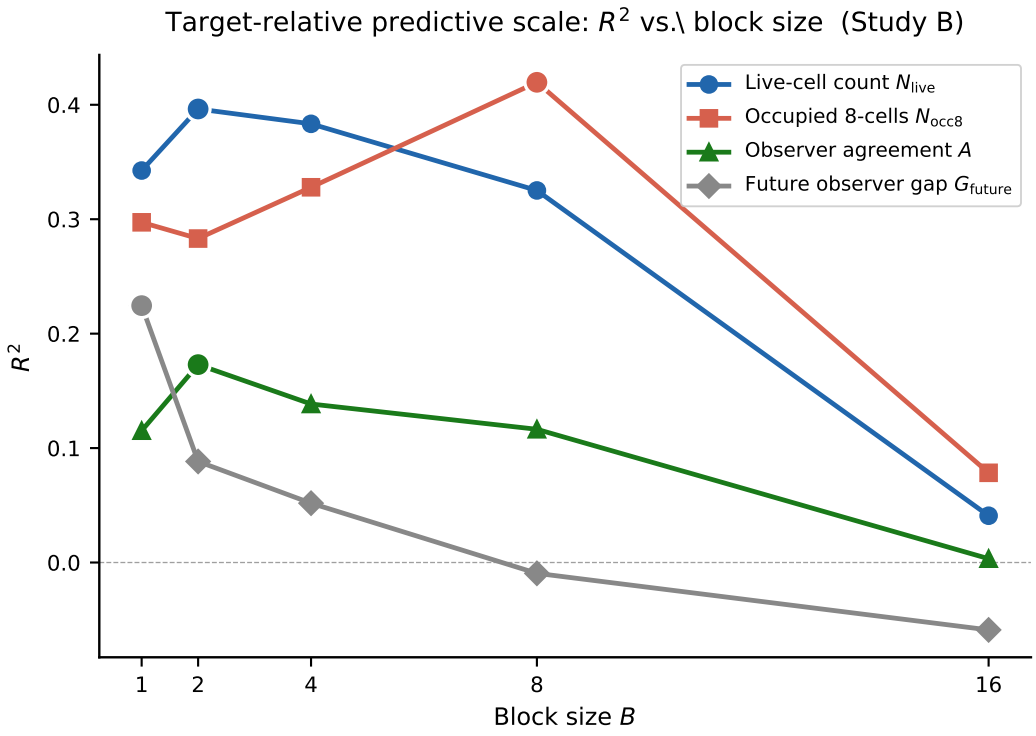

FIG. 11. **E3 — Target-relative predictive scale (Study B).** R^2 of a block-aggregated feature predictor as a function of block size B, separately for four prediction targets. N_{live} peaks at $B = 2$; N_{occ8} peaks at $B = 8$; future observer gap G_{future} peaks at $B = 1$ and falls steeply; observer agreement A is broadly flat then drops. Different targets favour different descriptive scales, motivating the target-specific framing of the main-text analysis. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

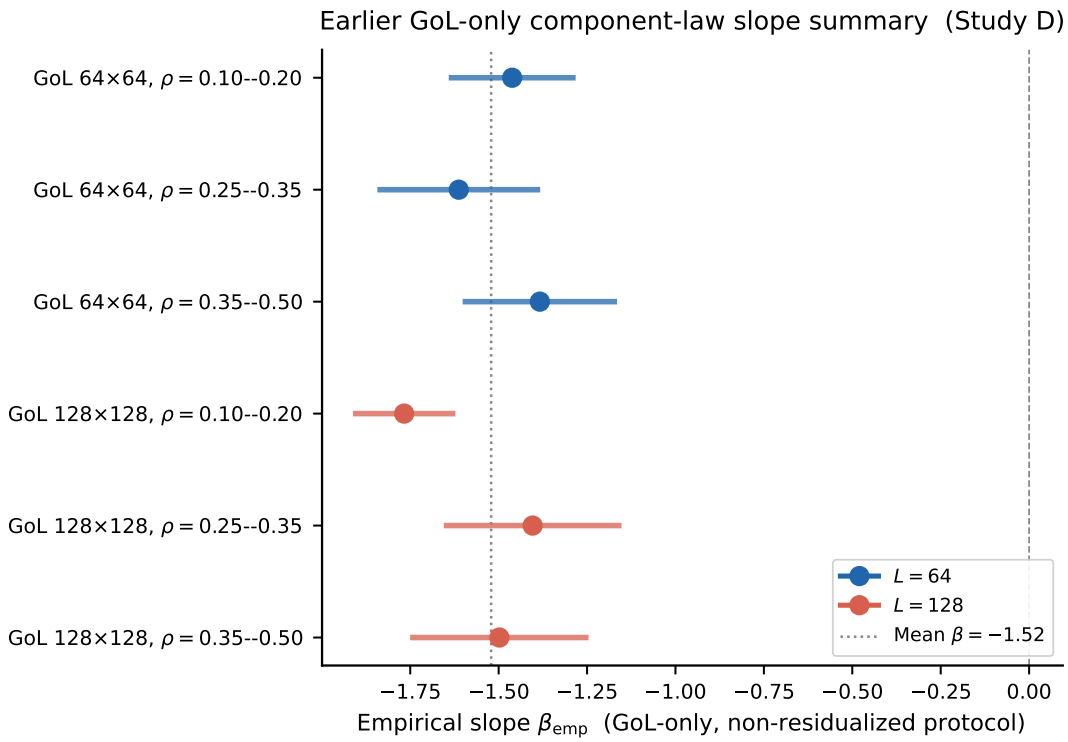

FIG. 12. **E4 — Earlier GoL-only component-response slope summary (Study D).** Empirical slopes β_{emp} with 95% CI for six GoL conditions (two grid sizes, three density bands) under an earlier, non-residualized regression protocol. Mean $\beta \approx -1.52$. **This coefficient is not directly comparable to the residualized slopes $\beta_{\text{iso}} \approx -0.70$ in the main text:** the two studies use different targets, different residualization, and different simulation batteries. This figure is included as historical lineage only. *Background diagnostic only; not load-bearing for the response estimates in the main text.*